

CHIARA VILLA

Languages Italian, English, French
Software MATLAB, Python, Stan (C++), Fortran90, COMSOL, Maple, R, HTML5
Memberships ESMTB, SMB, SMAI, MCAA, ComplexCité, EMS-TAG-MLS

ACADEMIC APPOINTMENTS

11/25 - *today* **Chargée de recherche CNRS**, MAP5, Université Paris Cité, Paris (FR)

Postdoctoral positions

02/25 - 08/25 **Postdoc**, Inria Saclay Centre, Palaiseau (FR)
Postdoctoral researcher in the Inria team MUSCA, project OvoTox/OVOPAUSE
01/23 - 12/24 **PRFP postdoc**, Laboratoire Jacques-Louis Lions, Sorbonne Université, Paris (FR)
Laureate of the Paris Region Fellowship Programme (PR and EU MSCA funding)
04/22 - 12/22 **Postdoc**, Laboratoire Jacques-Louis Lions, Sorbonne Université, Paris (FR)
Postdoctoral researcher in the group of Prof Benoît Perthame (ERC ADORA funding)

EDUCATION & RESEARCH EXPERIENCE

09/18 - 03/22 **PhD, Mathematics**, University of St Andrews, St Andrews (UK)
Supervisors: Prof Mark Chaplain, Dr Tommaso Lorenzi
Thesis: ‘Partial differential equation modelling in cancer and development’
Viva voce examiners: Prof Philip Maini (UniOx), Dr Nikolaos Sfakianakis (UoStA)
01/22 - 03/22 Research visit at Institute Henri Poincaré, Paris (FR)
2014 - 2018 **MMaths, Applied Mathematics**, University of St Andrews, St Andrews (UK)
Fast Track, First Class Honours awarded. Academic Prizes: Dean’s list (2014-2018),
The Principal’s Scholarship for Academic Excellence
Summer 2017 Undergraduate Summer Research Internship, University of St Andrews (UK)
Summer 2016 Complex Systems Biology Research Internship, Università degli Studi di Torino (IT)

SUPERVISION, TEACHING AND MARKING

Supervision of postgraduate students

01/26 - *today* **PhD thesis supervision** (co-encadrante) of Perla Mallouk (MAP5, UPCité)
09/23 - *today* **PhD thesis supervision** (co-encadrante) of Federica Padovano (LJLL, SU)
01/26 - 07/26 **Master thesis** (M2) of Tom Matrat (amU), at ESILV and MAP5 (UPCité)
01/23 - 08/23 **Master thesis** of Federica Padovano (EPFL), at LJLL (SU)

Teaching and Marking

All activities of 2017-2022 undertaken with the School of Mathematics and Statistics, University of St Andrews. Teaching activities undertaken with groups of 50 (demonstrating) or 11 (tutoring) students. Feedback on Explanation (E), Organisation (O) and Availability (A) on a scale of 1 (excellent) to 5 (poor).

Autumn 2020 **MT2000 Computing Workshop**, Demonstrator of computing in Python
Autumn 2019 **MT2000 Computing Workshop**, Demonstrator of computing in Python
Autumn 2019 **MT2501 Linear Mathematics**, Tutor of 2 groups (E=1.44, O=1.33, A=1.33)
Spring 2019 **MT2507 Mathematical Modelling**, Tutor of 2 groups (E=1.45, O=1.85, A=1.45)
Spring 2019 **MT2507 Mathematical Modelling**, Demonstrator of 3 groups
Autumn 2018 **MT2503 Multivariate Calculus**, Tutor of 2 groups (E=1.17, O=1.5, A=1.17)
Autumn 2018 **MT2504 Combinatorics and Probability**, Marking of 100 computing projects
Autumn 2017 **UK Undergraduate Ambassadors Scheme**, weekly teaching in secondary school (S1, S3, Advanced Higher Maths), *Waid Academy, Anstruther (UK)*

FUNDING, GRANTS AND PRIZES AWARDED

Competitive positions

- 2025 **CNRS Concours Chercheurs 2024**, ranked 1st at the concours de Chargé de recherche de classe normale n° 51/02 of the CNRS CDI51 (affiliation Insmi).
- 2025 **COMPETITION n°9 SIF-CRCN-2025 (INRIA)**, ranked 1st at the concours de Chargés de recherche de classe normale n°9-SIF-CRCN-2025 of the Centre INRIA de Saclay – *renounced to join the CNRS on 01/11/25*.
- 2025 **MSCA Postdoctoral Fellowships** (EU HORIZON-MSCA-2024-PF-01-01) with the project “Modelling of Epithelial-to-Mesenchymal Transition in Invading Cells” (score: 95.40%, **€260347.92**) – *renounced to join the CNRS on 01/11/25*.
- 2022 **Paris Region Fellowship Programme** (EU Horizon E2020, MSCA Grant Agreement no 945298-ParisRegionFP) with the proposed project on “Mechanistic models of cell migration and cancer invasion” (**€257760**).

Additional grants

- 2023 **PEPS JCJC** funding awarded by Insmi for the project “Conservative numerical schemes for novel structured PDE models of cancer invasion” with Alexandre Poulain (**€4900**).
- 2023 **BOUM SMAI funding**, MC2D workshop organisation, Paris 10/2023 (**€1000**).
- 2023 **UFR 929 funding**, Sorbonne Université, MC2D workshop organisation (**€1500**).
- 2021 **IHP financial support** for the “Mathematical modeling of organization in living matter” thematic program at the Institute Henri Poincaré during 10/01-01/04 2022 (**€4500**).

Smaller awards

- 2022 **Junior Fellowship** for the participation to the workshop “Parabolic and kinetic models in population dynamics” in Toulouse in September 2022.
- 2020 **SMBdevBio Poster Prize 1**, Society for Mathematical Biology, SMB2020 (**\$250**)
- 2020 **LMS ECR Travel Grant**, London Mathematical Society, 12th ECMTB (**£500**)
- 2018 **The Principal’s Scholarship for Academic Excellence**, prize awarded to the top 50 academically performing students in their final year at the UoStA (**£1000**)
- 2014 - 2018 **The Deans’ list**, annual award for academic excellence by the Deans of the UoStA
- 2017 **Research scholarship**, Undergraduate Summer Research Internship, UoStA (**£1684.29**)

PROFESSIONAL RESPONSIBILITIES

Journal peer reviewing (12 manuscripts)

PLOS Computational Biology, Journal of Mathematical Biology, International Journal of Non-Linear Mechanics, Bulletin of Mathematical Biology, iScience, European Journal of Applied Mathematics, Mathematical Biosciences, European Control Conference 2022, Frontiers in Ecology and Evolution

Mentoring and Representation

- 11/25 - today **Math C pour L** stage organiser, *MAP5, Université Paris Cité*
- 04/26 **Rendez-Vous des Jeunes Mathématiciennes et Informaticiennes**, *MAP5, Université Paris Cité*, Testimonial for female high school students interested in pursuing STEM subjects in higher education
- 04/26 **W MathBio!** invited speaker at the online seminar for women in Mathematical Biology
- 01/23 - 12/24 **Member of ‘Comité Parité’**, *Laboratoire Jacques-Louis Lions, Sorbonne Université*
- 10/22 - 12/24 **Postdoctoral Research Rep**, *Laboratoire Jacques-Louis Lions, Sorbonne Université*
- 09/18 - 06/22 **Mentor in Peer Mentoring scheme** of 4 Undergraduate, 3 Master, 2 PhD students
- 11/20 **Piscopia Society**, *School of Mathematics and Statistics, University of St Andrews*, Testimonial encouraging female/non-binary students considering a PhD in mathematics
- 09/18 - 09/19 **Postgraduate Research Rep & Postgraduate Research Executive Rep**, UoStA
- 09/18 - 09/19 **University of St Andrews Student Rep**, *SMSTC*

SELECTED SCIENTIFIC MEETINGS

Scientific meetings organised

- Jul 2026 **Minisymposium** ‘Identifying the origin and consequences of non-genetic cell-to-cell variability’, *ECMTB2026*
- Jun 2025 **Minisymposium** ‘Mathematical advances in modelling cancer treatment’, *SMAI 2025*
- Jul 2024 **Minisymposium** ‘Recent advances in modelling cancer invasion’, *ECMTB2024*
- Oct 2023 **Workshop** ‘Mathematical challenges in modelling cancer dynamics’ (~ 50 ppl), *Laboratoire Jacques-Louis Lions, Sorbonne Université*, <https://mc2d.sciencesconf.org/>
- 2020-2021 **StAMBio seminar series**, Weekly talks (online) by members of the St Andrews Mathematical Biology research group and international guest speakers
- Jan 2020 **Postgraduate Interdisciplinary Mathematics Symposium**, for PhD students of the School of Mathematics and Statistics of the University of St Andrews, *Edzell*
- Dec 2019 **Workshop** ‘Scottish Mathematical Biology Forum’, *ICMS, Edinburgh*

Invitations to speak at International Workshops

- Sep 2026 **Kinetic and Macroscopic models in Biology**, *Centre de Recerca Matemàtica*
- Jul 2026 **Mathematical modeling in biology and medicine**, *Wolfgang Pauli Institute*
- Jun 2026 **In silico Health**, *DVRS, ESILV and Centre Hospitalier de Saint-Denis*
- Feb 2026 **From Micro to Macro in Mathematical Biology**, *Politecnico di Torino*
- Dec 2025 **New trends in mathematical models for biology**, *Institut Henri Poincaré*
- Apr 2025 **Young women in Mathematical Biology**, *Bonn University*
- Apr 2024 **Mathematical and numerical tools for Oncology**, *Oncolille Institut*
- Nov 2023 **Mechanistic models for continuous phenotypic adaptation**, *University of Leeds*
- Jun 2023 **Mathematical Biology: Analysis and Application**, *Technische Universität Dresden*
- Feb 2023 **Multiscale analysis and methods for PDEs**, *Institute for Mathematical Sciences (SG)*
- Oct 2022 **Modelling cell and tissue biomechanics**, *LJLL, Sorbonne University*
- Jun 2021 **Soft Tissue Mechanics**, *University of St Andrews, Online*
- May 2021 **Mathematical Biology on the Mediterranean Coast**, *LJLL, SU, Online*
- Jun 2020 **Interplay between Oncology, Mathematics and Numerics**, *LJLL, Inserm, Online*

Invitations to speak at Conference Mini-symposia:

- Jul 2026 **14th ECMTB**, ‘Modeling collective dynamics in heterogeneous cell populations’, *Graz*
- Jul 2024 **13th ECMTB**, ‘Modelling heterogeneity, adaptation and evolution in cancer’, *Toledo*
- Jul 2024 **ECM**, ‘Mathematical methods for biological problems’, *Sevilla*
- Aug 2023 **SIMAI**, ‘Models and methods for biomedical applications’, *Matera*
- Sep 2022 **12th ECMTB**, ‘Evolutionary dynamics in structured populations’, *Heidelberg*

Invitations to speak at Seminars:

- Nov 2024 **Séminaire de modélisation mathématique en sciences de la vie et santé**, *LJLL*
- May 2024 **Mathematical Biology Seminar**, *University of Leeds*
- Apr 2024 **Puissant Lab**, *Saint-Louis Medical Center (Paris)*
- May 2023 **The Evolution Seminar**, *Bielefeld University*
- Apr 2023 **Analyse Numérique et Équations aux Dérivées Partielles**, *Université de Lille*
- Nov 2022 **Synthsys Seminar**, *Centre for Synthetic and Systems Biology, Edinburgh*

PUBLICATIONS

- [15] F. Padovano, A.M. Lombardi, S. Novello, T. Lorenzi, E. Vigna, E. Leschiera, C. Villa, Modelling the efficacy of CAR-CIK cell immunotherapy targeting MET-expressing mesothelioma cells, *Accepted in Scientific Reports (In press – preprint on ResearchGate)*, 2026.
- [14] B. Perthame, C. Villa, Existence, regularity and stability in a strongly degenerate nonlinear diffusion and haptotaxis model of cancer invasion, *SIAM Journal on Mathematical Analysis*, 58(3):2948-2964, 2026. DOI: 10.1137/24M1722079, **hal-04854773**
- [13] T. Lorenzi, N. Loy, C. Villa, Phenotype-structuring of non-local kinetic models of cell migration driven by environmental sensing, *SIAM Multiscale Modeling & Simulation*, 24(3):1143-1172, 2026. DOI: 10.1137/24M1719566, **hal-04851469**
- [12] S. Hamis, A.P. Browning, A.L. Jenner, C. Villa, P. K. Maini and T. Cassidy, Growth rate-driven modelling reveals how phenotypic adaptation drives drug resistance in BRAFV600E-mutant melanoma, *Communications Biology*, 9(1):385, 2026. DOI: 10.1038/s42003-026-09760-2, **hal-04851795**
- [11] L. Almeida, A. Poulain, A. Pourtier, C. Villa, Mathematical modelling of the contribution of senescent fibroblasts to basement membrane digestion during carcinoma invasion, *Mathematical Modelling of Natural Phenomena*, 21(1):11, 2026. DOI: 10.1051/mmnp/2026002, **hal-04574340**
- [10] T. Lorenzi, K.J. Painter, C. Villa, Phenotype structuring in collective cell migration: a tutorial into mathematical models and methods, *Journal of Mathematical Biology*, 91(2):22, 2025. DOI: 10.1007/s00285-025-02246-5, **hal-04851615**
- [9] A.P. Browning, R. Crossley, C. Villa, P. K. Maini, A.L. Jenner, T. Cassidy and S. Hamis, Identifiability of heterogeneous phenotype adaptation from low-cell-count experiments and a stochastic model, *PLOS Computational Biology*, 21(6):e1013202, 2025. DOI: 10.1371/journal.pcbi.1013202, **hal-04854906**
- [8] C. Villa, P. K. Maini, A.P. Browning, A.L. Jenner, S. Hamis and T. Cassidy, Reducing phenotype-structured PDE models of cancer evolution to systems of ODEs: a generalised moment dynamics approach, *Journal of Mathematical Biology*, 91(2):22, 2025. DOI: 10.1007/s00285-025-02246-5, **hal-04599519**
- [7] F. Padovano, C. Villa, The development of drug resistance in metastatic tumours under chemotherapy: an evolutionary perspective, *Journal of Theoretical Biology*, 595(1):111957, 2024. DOI: 10.1016/j.jtbi.2024.111957, **hal-04595087v3**
- [6] L. Almeida, J.A. Denis, N. Ferrand, T. Lorenzi, M. Sabbah, C. Villa, Evolutionary dynamics of glucose-deprived cancer cells: insights from experimentally-informed mathematical modelling, *Journal of the Royal Society Interface*, 21(210):20230587, 2024. DOI: 10.1098/rsif.2023.0587, **hal-03947209v2**.
- [5] C. Villa, A. Gerisch, M.A.J. Chaplain, A novel nonlocal partial differential equation model of endothelial progenitor cell cluster formation during the early stages of vasculogenesis, *Journal of Theoretical Biology*, 534(1):110963, 2022. DOI: 10.1016/j.jtbi.2021.110963, **hal-04415625**.
- [4] F. Mottes, C. Villa, M. Osella, M. Caselle, The impact of whole genome duplications on the human gene regulatory networks, *PLOS Computational Biology*, 17(12):e1009638, 2021. DOI: 10.1371/journal.pcbi.1009638 **hal-04415666**.
- [3] C. Villa, M.A.J. Chaplain, A. Gerisch, T. Lorenzi, Mechanical models of pattern and form in biological tissues: the role of stress-strain constitutive equations, *Bulletin of Mathematical Biology*, 83:80, 2021. DOI: 10.1007/s11538-021-00912-5, **hal-04415645**.
- [2] C. Villa, M.A.J. Chaplain, T. Lorenzi, Evolutionary dynamics in vascularised tumours under chemotherapy: Mathematical modelling, asymptotic analysis and numerical simulations, *Vietnam Journal of Mathematics*, 49, 143–167, 2021. DOI: 10.1007/s10013-020-00445-9, **hal-04415601**.
- [1] C. Villa, M.A.J. Chaplain, T. Lorenzi, Modelling phenotypic heterogeneity in vascularised tumours, *SIAM Journal on Applied Mathematics*, 81, 434–453, 2021. DOI: 10.1137/19M1293971, **hal-04415631**.